INORGANIC CUME

December 4, 1993

Metal Carbonyls

18-1. Give the formulas and sketch the structures for three different neutral mono-nuclear (mono-metal) carbonyls which each have a different point group molecular symmetry. Give the point group of each.

18-2. Give the formulas and sketch (showing all bonds) three different neutral carbonyls which each contain two metal atoms. The three should differ in structure or stoichiometry.

18-3. Sketch two isomers possible for the tri-iron carbonyl.

18-4. Give the formula and sketch the structure for any carbonyl with four metal atoms.

18-5. For the two complexes $M(CO)_5L$ and $M(CO)_5L'$ assume L and L' are both phosphines and L' forms a better pi bond with the metal than does L. How would you show this relative pi bonding of L and L' experimentally? (Give the details of the experiment, the expected results, and a full explanation for the results)

18-6. Use a molecular orbital explanation to show the effect of pi bonding by CO on the energy gap of the crystal field splitting in an octahedral complex.

18-7. Give a reasonable series of reactions for the preparation of nickel carbonyl starting with nickel chloride.

18-8. Give a reasonable reaction scheme for preparing $M(CO)_5L$ from the hexacarbonyl in such a way that no $M(CO)_4L_2$ will be formed.